Lab 9 – Frequency Spectrum

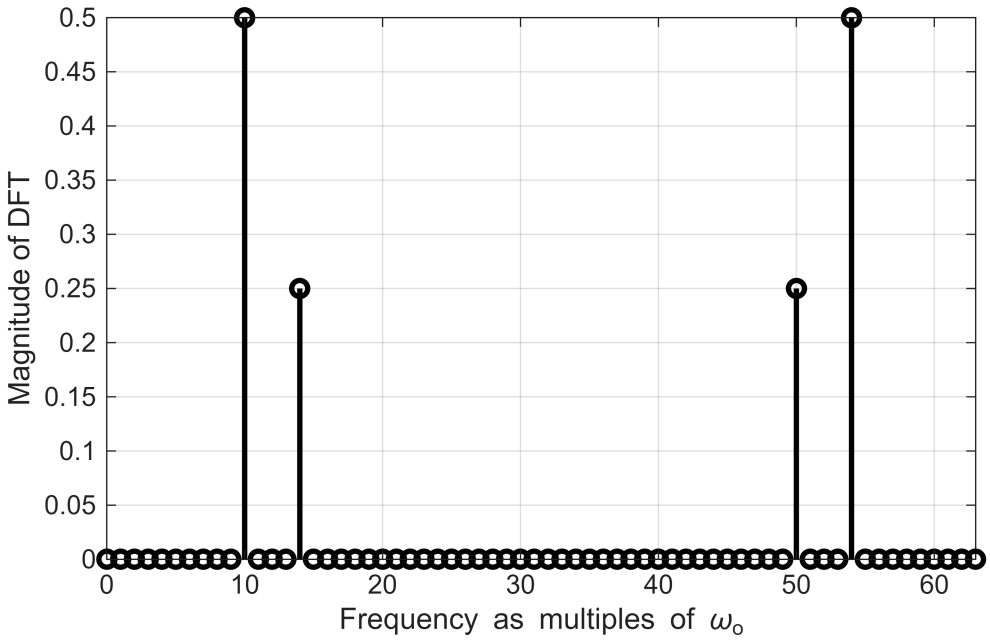
EE252L/251L: Signals and Systems Lab

Dr. Basit Memon and Ms. Hira Mustafa

Name: M. Shees Abbas **ID: ma11212**

Task 1:

```
P = 64;
w = 2*pi/P;
n = 0:P-1;
x = cos(10*w*n) + 0.5*cos(14*w*n); %Finite-time signal
hx = fft(x); %Compute the DFT of x
stem(n,abs(hx)/P,'k','LineWidth',2); %Compute and plot magnitude of coefficients
ylabel('Magnitude of DFT');
xlabel('Frequency as multiples of \omega_o');
axis([0 P-1 0 inf]);
grid;
```



Task 2:

(a)

```
clc; clear; close all;
fs = 8000;
n = 1000;
```

```
t = (0:n-1)/fs;
w1 = 400;
x1 = sin(2*pi*w1*t);
```

(b)

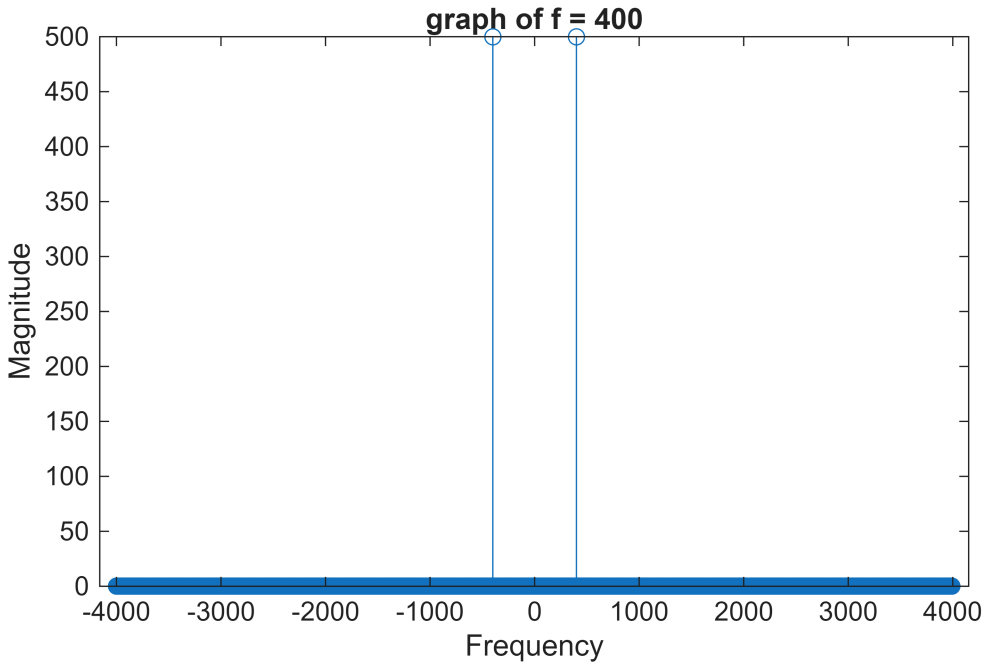
```
hx1 = fft(x1,n);
hx1_shifted = fftshift(hx1);
```

(c)

```
fhz = (-n/2:(n/2)-1)*(fs/n);
```

(d)

```
figure;
stem(fhz, abs(hx1_shifted))
title('graph of f = 400')
ylabel('Magnitude');
xlabel('Frequency');
```

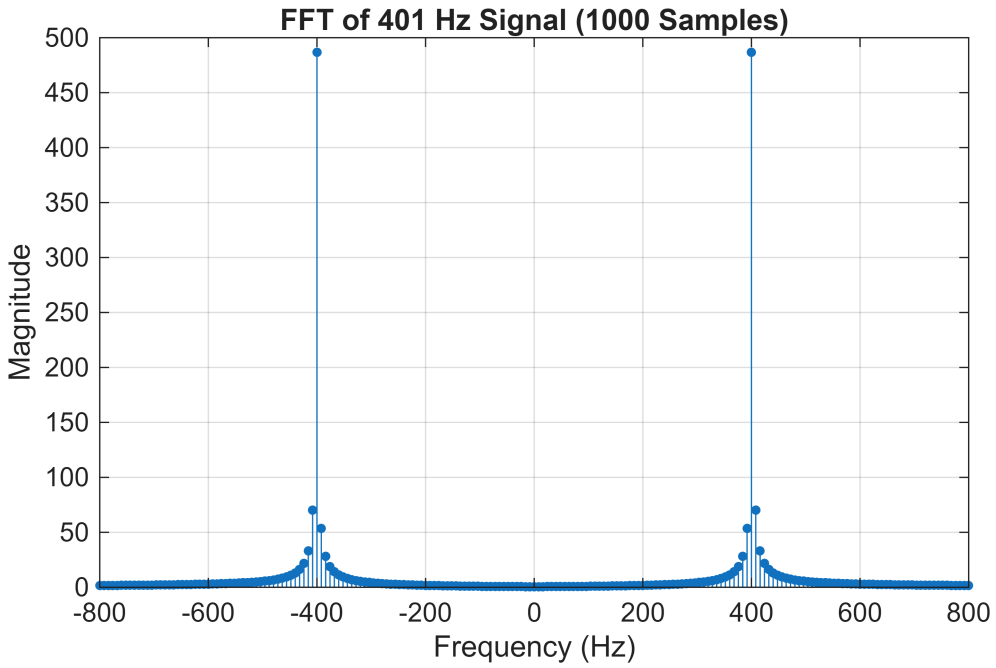


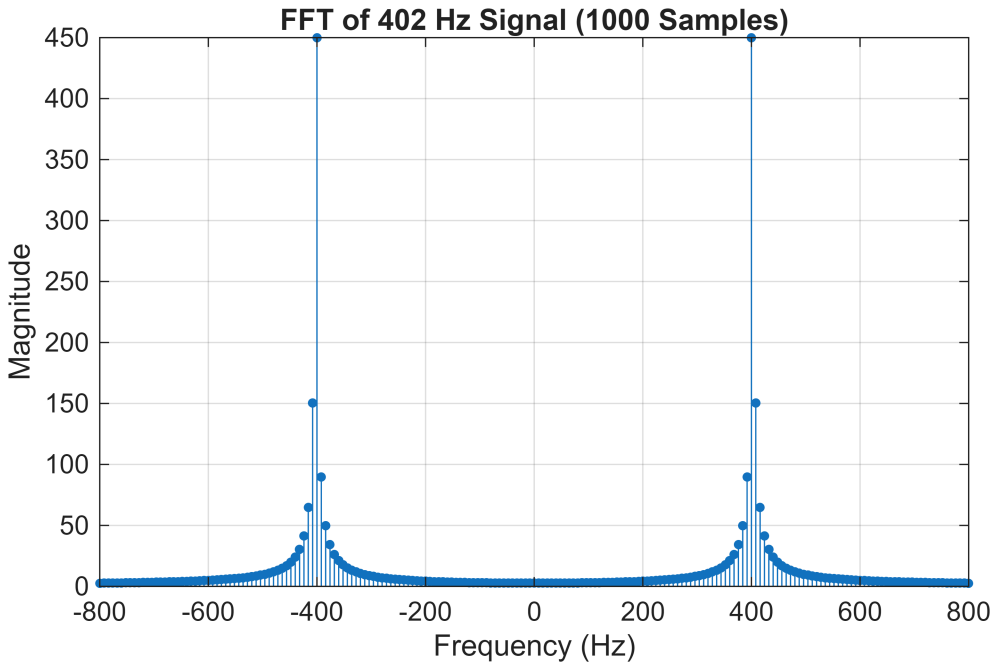
(e)

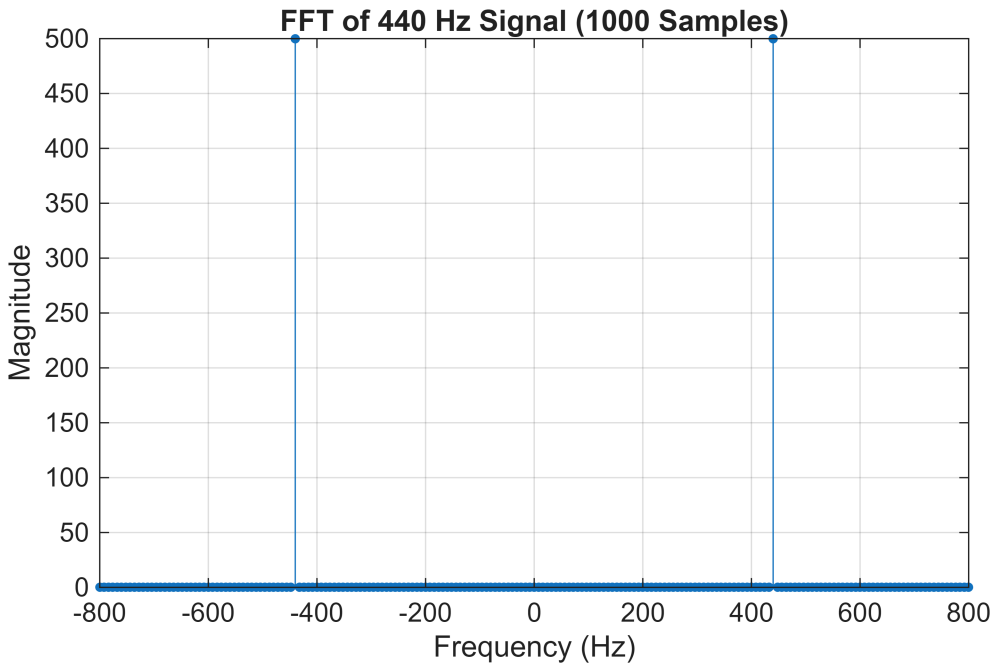
```
frequencies = [401, 402, 440];
for i = 1:length(frequencies)
    f = frequencies(i);
    x = sin(2 * pi * f * t);
    hx = fftshift(fft(x, n));

    figure;
    stem(fhz, abs(hx), 'filled', 'MarkerSize', 3);
    title(['FFT of ', num2str(f), ' Hz Signal (1000 Samples)']);
```

```
    xlabel('Frequency (Hz)');
    ylabel('Magnitude');
    grid on;
    xlim([-800 800]);
end
```







400 Hz and 440 Hz both frequencies result in an **integer number of complete cycles** over the 1000-sample window, the waveform starts and ends at the exact same phase point seamlessly. There are no sudden truncation jumps at the edges of the 1000-sample block.

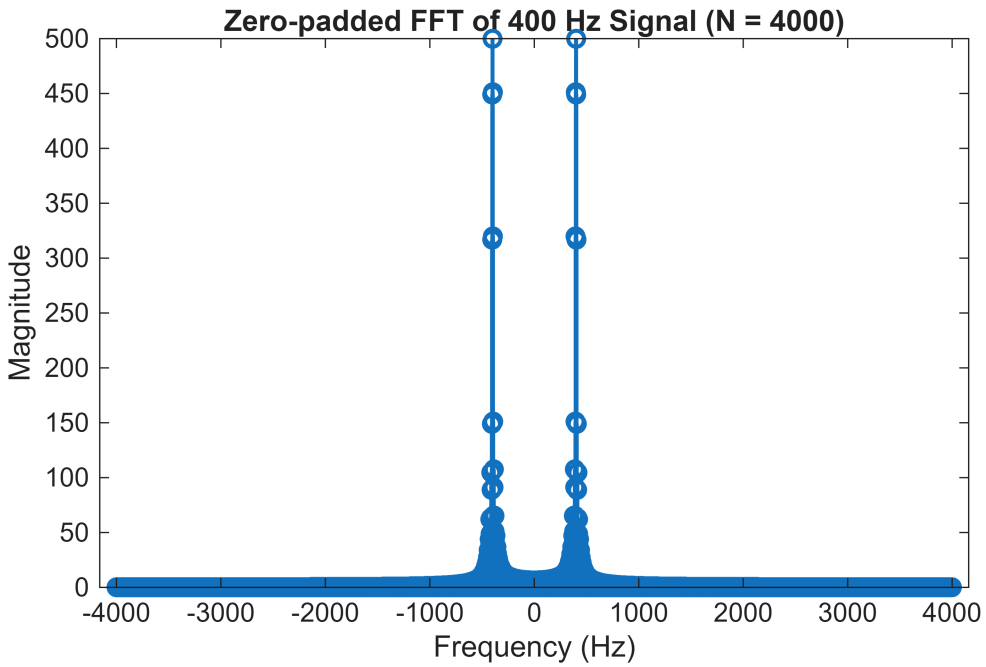
In contrast, a frequency like **401 Hz** yields 50.125 cycles. The window cuts the sine wave mid-cycle, causing a harsh discontinuity between the start and end of the data block. This sharp edge creates **spectral leakage**, spreading the energy into neighboring frequency bins.

(f)

```
fs = 8000;
n_samples = 1000;
t = (0:n_samples-1)/fs;
w1 = 400;
x1 = sin(2*pi*w1*t);

N = 4000;
hx2 = fft(x1, N);
hx2_shifted = fftshift(hx2);
fhz = (-N/2 : (N/2)-1) * (fs / N);

figure;
stem(fhz, abs(hx2_shifted), 'LineWidth', 1.5);
title('Zero-padded FFT of 400 Hz Signal (N = 4000)');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
```



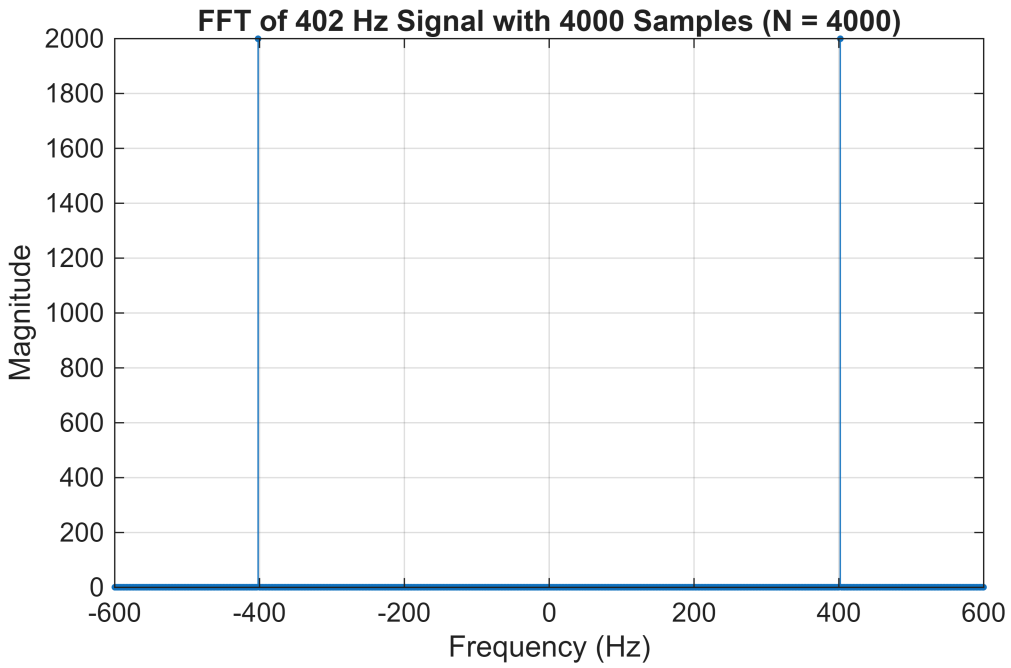
The graph will change from discrete stems into a **smooth, continuous curve**. Because zero-padding interpolates the frequency bins, it fills in the gaps between the original points to outline the shape of the spectral leakage envelope, making the overall curve look continuous.

(g)

```
n_long = 4000;
t_long = (0:n_long-1)/fs;
w_402 = 402;
x_402_long = sin(2*pi*w_402*t_long);

hx_402_long = fft(x_402_long, n_long);
hx_402_long_shifted = fftshift(hx_402_long);
fhz_long = (-n_long/2 : (n_long/2)-1) * (fs / n_long);

figure;
stem(fhz_long, abs(hx_402_long_shifted), 'filled', 'MarkerSize', 2);
title('FFT of 402 Hz Signal with 4000 Samples (N = 4000)');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
grid on;
xlim([-600 600]);
```



The graph will display a **much taller, sharper, and narrower peak** at 402 Hz. The messy, widespread "side lobes" (spectral leakage) from the 1000-sample version will disappear, leaving a clean, isolated single-frequency spike.

Task 3

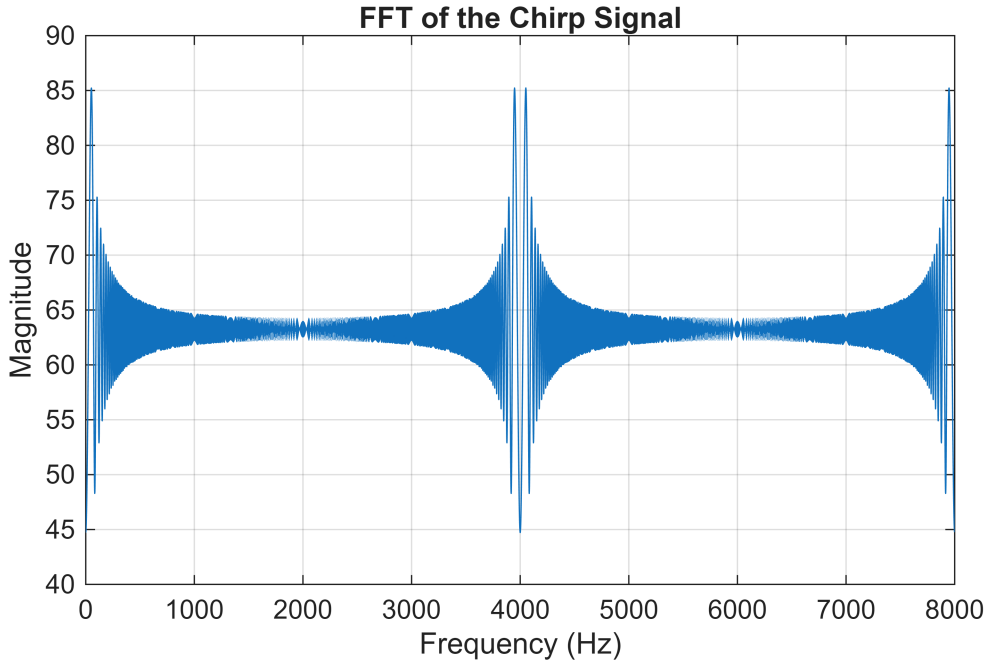
(a)

```
fs = 8000;
N = 8000;
t = (0:N-1) / fs;
x = chirp(t,0,1,4000);
```

(b)

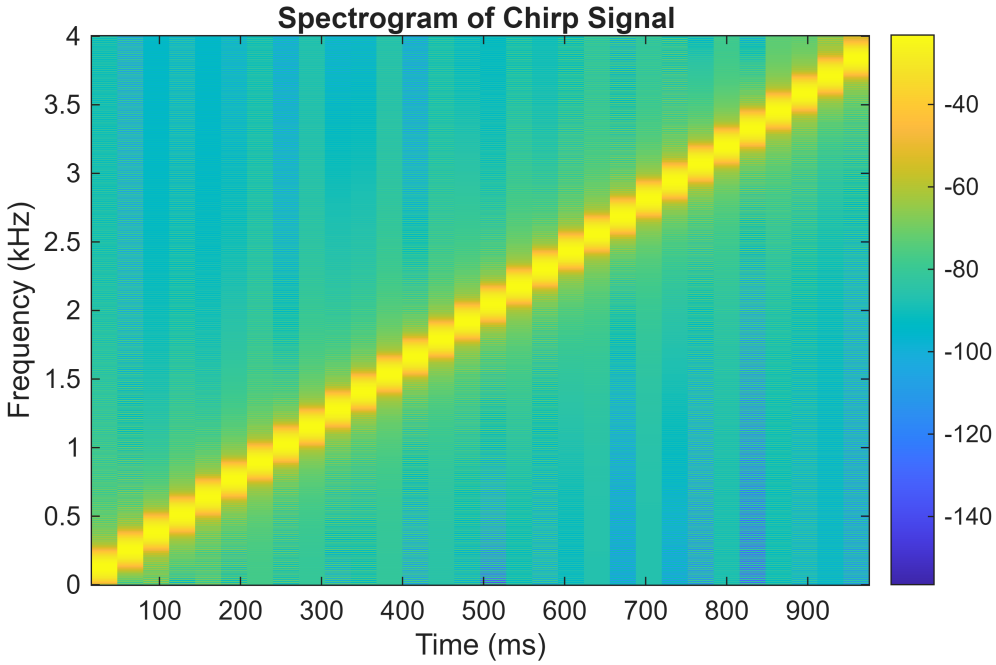
```
X = fft(x);
f_axis = (0:N-1) * (fs / N);

figure;
plot(f_axis, abs(X));
title('FFT of the Chirp Signal');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
grid on;
```



(c)

```
figure;
spectrogram(x, 512, [], 1024, fs, 'yaxis');
title('Spectrogram of Chirp Signal ');
colorbar;
```

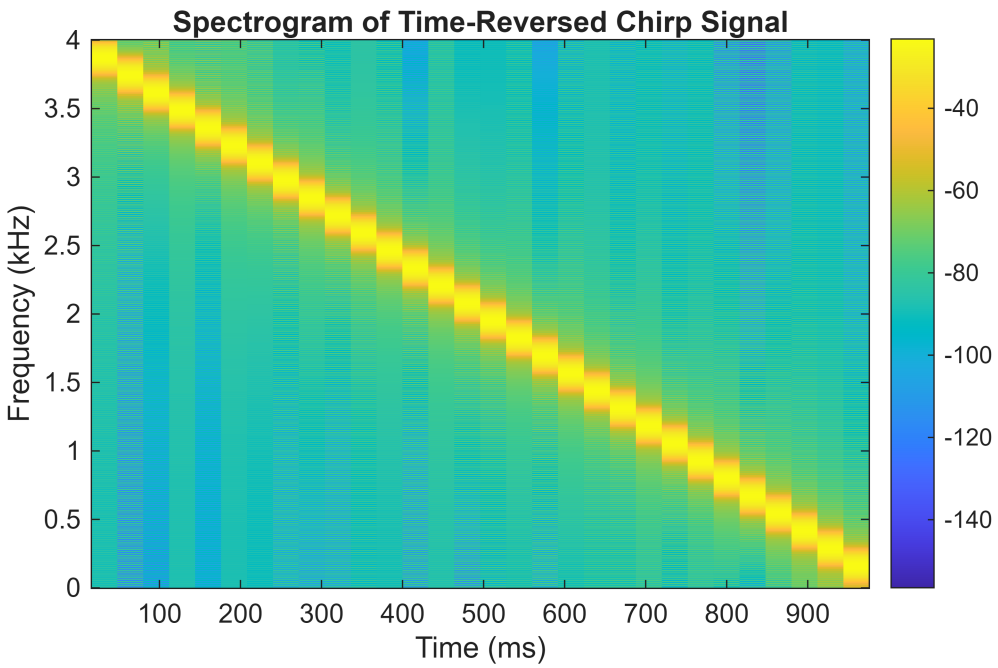


the resulting spectrogram displays a clear diagonal line sloping upwards from **0th sample** to **8000 sample**, which perfectly agrees with our understanding and theoretical derivation of the signal's time-varying frequency. Over the 1-second duration, the frequency increases linearly

(d)

```
x_rev = x(8000:-1:1);

figure;
spectrogram(x_rev, 512, [], 1024, fs, 'yaxis');
title('Spectrogram of Time-Reversed Chirp Signal');
colorbar;
```



This makes complete sense because time-reversing the signal (x(-t)) reverses the order in which the frequency components are presented. Instead of starting at **0th sample** and sweeping up to **8000th sample**, the reversed signal starts at **8000th sample** and sweeps downwards to **0th sample**, which is represented on the spectrogram by a diagonal line sloping downwards from left to right

Assessment Rubric

Lab 9: Frequency Spectrum

Name:	**Student ID:**

Marks distribution:

	Tasks	**LR2 Code**	**LR5 Results**	**LR9 Report**	**AR4 Report Submission**	**AR7 Comments**
In-Lab	**Task 1**	/2	/4	/6	/12	
	Task 2	/8	/16	/8		/2
	Task 3	/4	/8	/8		/2
Total = 80		/14	/28	/22	/12	/4
CLO Mapped		**CLO2**	**CLO2**	**CLO2**	**CLO2**	**CLO2**

CLO – (LDL)	**Total Points**	**Points Obtained**
CLO2 – (Psy-4)	80	
Total	**80**	

For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.

Lab Evaluation Assessment Rubric

#	Assessment Elements	Level 1: Unsatisfactory Points 0-1	Level 2: Developing Points 2	Level 3:Good Points 3	Level 4:Exemplary Points 4
LR2	**Program/Code/ Simulation Model/ Network Model**	Program/code/simulation model/network model does not implement the required functionality and has several errors. The student is not able to utilize even the basic tools of the software.	Program/code/simulation model/network model has some errors and does not produce completely accurate results. Student has limited command on the basic tools of the software.	Program/code/simulation model/network model gives correct output but not efficiently implemented or implemented by computationally complex routine.	Program/code/simulation /network model is efficiently implemented and gives correct output. Student has full command on the basic tools of the software.
LR3	**Troubleshooting**	Unable to identify the fault/minimal effort show in troubleshooting.	Able to identify the fault but unable to remove it.	Able to identify the fault but partially removes it.	Able to identify the fault and takes necessary steps and actions to correct it.
LR4	**Data Collection**	Measurements are incomplete, inaccurate and imprecise. Observations are incomplete or not included. Symbols, units and significant figures are not included.	Measurements are somewhat inaccurate and imprecise. Observations are incomplete or vague. Major errors are there in using symbols, units and significant digits.	Measurements are mostly accurate. Observations are generally complete. Minor errors are present in using symbols, units and significant digits.	Measurements are both accurate and precise. Data collection is systematic. Observations are very thorough and include appropriate symbols, units and significant digits and task completed in due time.
LR5	**Results & Plots**	Figures/ graphs / tables are not developed or are poorly constructed with erroneous results. Titles, captions, units are not mentioned. Data is presented in an obscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captions and proper units are mentioned. Data presentation is systematic.
LR6	**Calculations**	Formulae used and/or calculations are incorrect. Units are not mentioned with results.	Formulae used are correct but there are few mistakes in calculation which lead to incorrect results.	Formulae used and end results are correct. Complete working steps are not shown. Proper units are not mentioned with results.	Formulae used, end results and all calculations are correct with all the intermediate steps clearly shown. Units are mentioned with results.
LR9	**Report**	All the in-lab tasks are not included in report and / or the report is submitted too late.	Most of the tasks are included in report but are not well explained. All the necessary figures / plots are not included. Report is submitted after due date.	Good summary of most the in-lab tasks is included in report. The work is supported by figures and plots with explanations. The report is submitted timely.	Detailed summary of the in-lab tasks is provided. All tasks are included and explained well. Data is presented clearly including all the necessary figures, plots and tables.
LR11	**Design**	Proposed design is unsubstantiated or does not satisfy most of given constraints. The breakdown of system into features is incomplete and still at lower resolution.	Justification for proposed design is weak, and it only satisfies some constraints. The breakdown of system is missing some features and resolution is not appropriate at some places.	Proposed design is substantiated, preferably through proper analysis, and satisfies most given constraints. The breakdown of system into features seems complete, but resolution is not appropriate at some places.	Proposed design is substantiated, preferably through proper analysis, and satisfies all given constraints. The breakdown of system into features seems complete and at appropriate resolution, given students' level.
AR4	***Report Submission**	Not accepted after 1 week.	Late submission after 2 days and within 1 week.	Late submission after the lab timing and within 2 days of the due date.	Timely submission of the report and in the lab time.
AR7	**Report Content/Code comments**	Most of the questions are not answered / figures are not labelled/ titles are not mentioned / units are not mentioned. No comments are present in the code.	Some of the questions are answered, figures are labelled, titles are mentioned and units are mentioned. Few comments are stated in the code.	Majority of the questions are answered, figures are labelled, titles are mentioned and units are mentioned. Comments are stated in the code.	All the questions are answered, figures are labelled, titles are mentioned and units are properly mentioned. Proper comments are stated in the code.